ECE 374 B - Summer 2026

# Homework 2

- **Submit your solutions electronically on the course Gradescope site as PDF files.** If you plan to typeset your solutions, please use the LaTeX solution template on the course web site. If you must submit scanned handwritten solutions, please use a black pen on blank white paper and a high-quality scanner app (or an actual scanner, not just a phone camera). We will mark difficult to read solutions as incorrect and move on.
- **Every homework problem must be done *individually*.** Each problem needs to be submitted to Gradescope before midnight of the due data which can be found on the course website: https://ecealgo.com/su26/homeworks.html.
- For nearly every problem, **we have covered all the requisite knowledge required to complete a homework assignment prior to the "assigned" date.** This means that there is no reason not to begin a homework assignment as soon as it is assigned. Starting a problem the night before it is due a recipe for failure.

## Policies to keep in mind

- **You may use any source at your disposal**—paper, electronic, or human—but you ***must*** cite ***every*** source that you use, and you ***must*** write everything yourself in your own words. See the academic integrity policies on the course web site for more details.
- **Being able to clearly and concisely explain your solution is a part of the grade you will receive.** Before submitting a solution ask yourself, if you were reading the solution without having seen it before, would you be able to understand it within two minutes? If not, you need to edit. Images and flow-charts are very useful for concisely explain difficult concepts.

**See the course web site (https://ecealgo.com/su26) for more information.**

If you have any questions about these policies,
please don't hesitate to ask in class, in office hours, or on Piazza.

1. For each scenario below, construct (or give a formal outline of) a nondeterministic finite-state automata that accepts the described set of strings. Keep the number of nodes in your machine minimal. If useful, include a brief rationale for your construction.

   (a) All strings over $\{a, b, c\}^*$ in which every nonempty maximal substring of consecutive $a$'s is of even length.

   (b) $\Sigma^* a \Sigma^* b \Sigma^* c \Sigma^*$

   (c) All strings in $w \in a^*$ of length that is divisible by at least one of the following numbers 2, 3, 5. For full credit your automata should have less than (say) 15 states.

   (d) All strings in $w \in a^*$ of length that is **NOT** divisible by at least one of the following numbers 2, 3, 5.

2. Suppose you are given a deterministic machine $M = (\Sigma, Q, \delta, s, A)$ that recognizes some language $L$. Your task is to design a nondeterministic counterpart $N$ that accepts a modified version of this language in each case below. Include a concise justification for your approach where relevant.

   (a) $\mathit{RemoveOnes}(L) := \{0^{\#_0(w)} \mid w \in L\}$; i.e., removes all 1s from the strings.

   (b) $\mathit{RemoveOnes}^{-1}(L) := \{w \in \Sigma^* \mid 0^{\#_0(w)} \in L\}$; i.e., puts back the 1s.

3. Use the fooling sets to show that each of the following languages is irregular.

   (a) $L = \{ww^R w \mid w \in \{0,1\}^*\}$

   (b) $L = \{0^i 1 0^j \mid i \text{ is divisible by } j\}$

   (c) $L = \{a^i b^j \mid j = \log_2 i\}$

   (d) $L = \{0^i 0^j \mid j = \sqrt{i}\}$

   (e) $L = \{wcd\#a(w) \mid w \in \{a, b\}^*\}$